

Hidden Markov Model (HMM)

Qian Wang

School of Biomedical Engineering

ShanghaiTech University

From Static to Dynamic

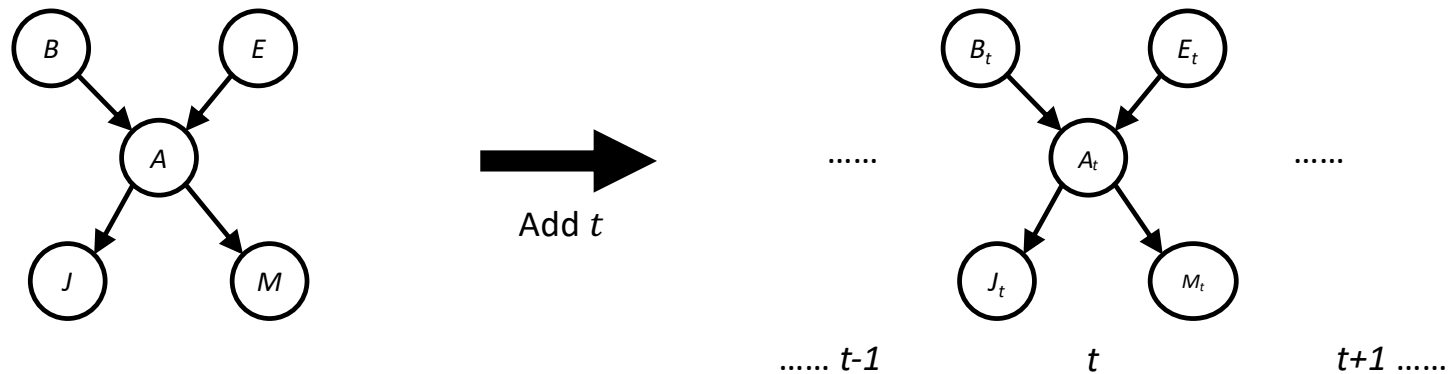
- We've developed strong capabilities for probabilistic reasoning in static environments
 - Example: Bayesian networks
- But real problem is often dynamic!
 - **Infer the current state** from a history of observations;
 - **Predict future outcomes** based on past and present evidence.
- Why static models fall short?
 - Static models don't capture temporal dependencies;
 - We need to **explicitly model time** in our systems.

Reasoning over Time (or Space)

- Often, we need to reason about a sequence of observations, such as
 - Speech recognition
 - Robot localization
 - User attention tracking
 - Medical monitoring
- To handle these tasks effectively, we must introduce **time (or space)** into our models.

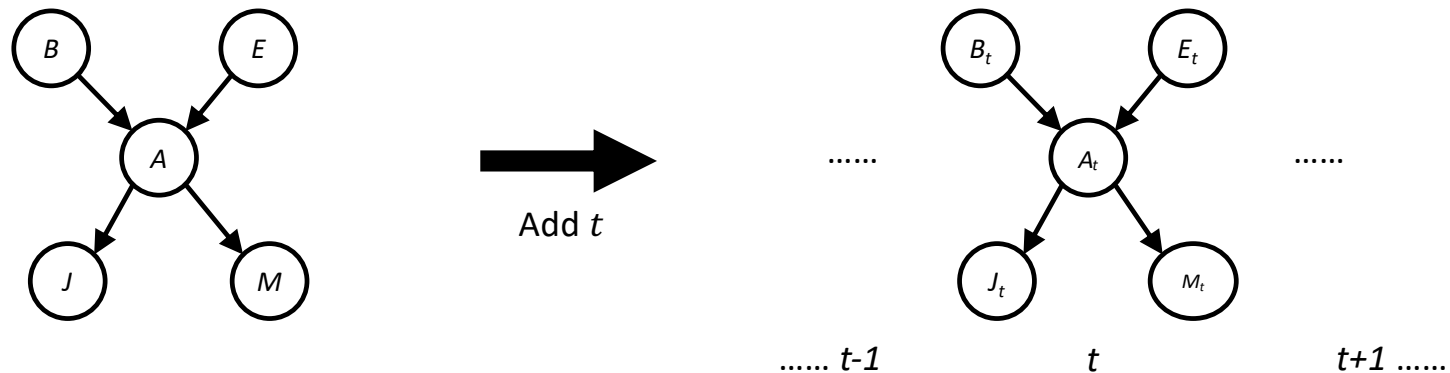
Discretization of t

- The world is **discretized** into a sequence of time slices;
- Each time slice contains a set of random variables, some of which are observable, others hidden;
- The same subset of variables is observable at each time step.



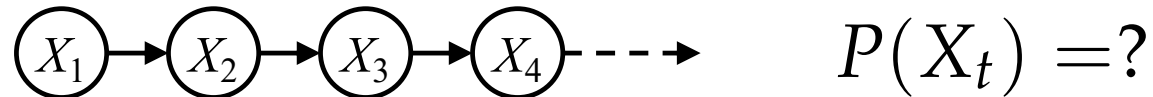
Discretization of t

- X_t : the set of **state variables** at time t , which are assumed to be hidden (unobservable).
- E_t : the set of **evidence variables** at time t , which are assumed to be observable.



Markov Chain

- **State:** Value of X at a given time is called the state.



- **Transition probabilities (dynamics):** $P(X_t | X_0 : X_{t-1})$ specifies how the state evolves over time t .
 - What will happen to $X_0 : X_{t-1}$, if t increases all the way?

Example Markov Chain: Weather

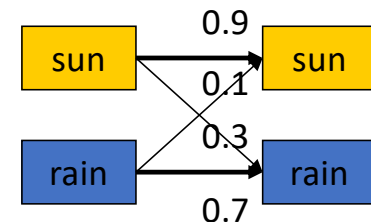
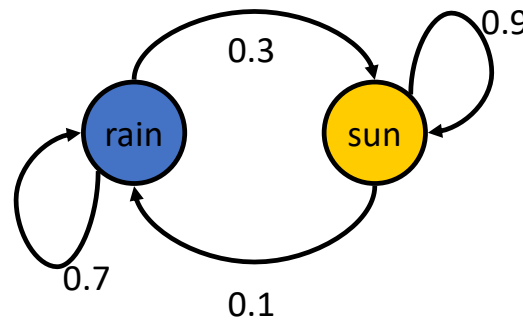
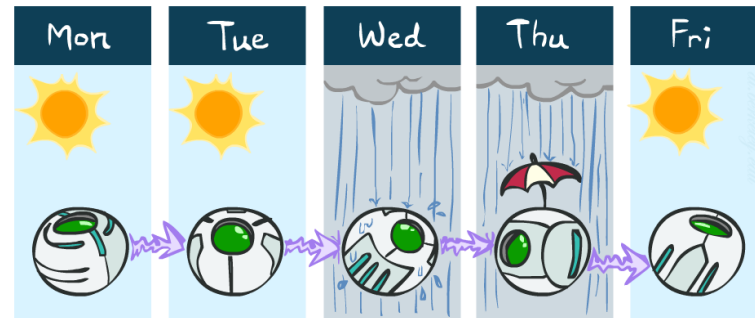
- State: $X = \{\text{rain}, \text{sun}\}$

Initial distribution:

$P(X_0)$	
sun	rain
1	0.0

CPT for $P(X_t | X_{t-1})$:

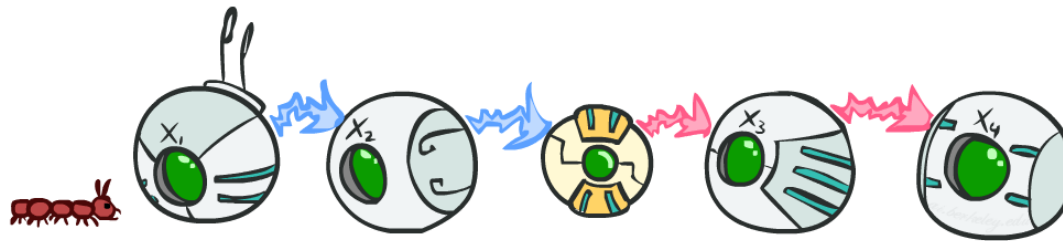
X_{t-1}	X_t	$P(X_t X_{t-1})$
sun	sun	0.9
sun	rain	0.1
rain	sun	0.3
rain	rain	0.7



Two new ways of representing the same CPT

Markov Property

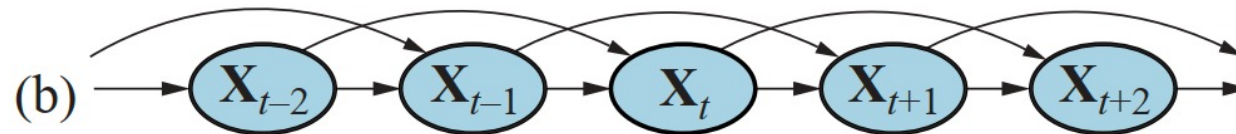
- Markov assumption: The current state depends only on a finite, fixed number of previous states;
- Any process that satisfies this assumption is called a **Markov process** or a **Markov chain**.



First- and Second-order Markov Chains

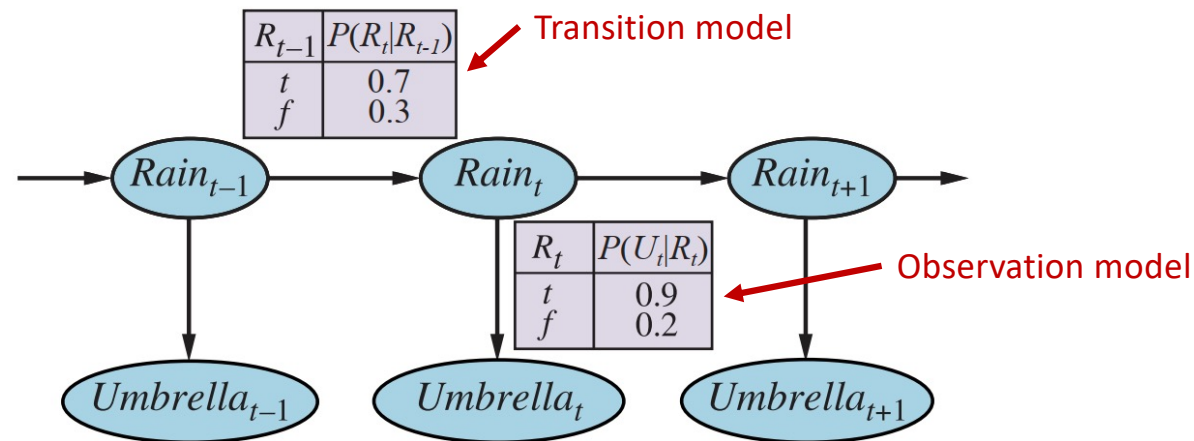
- Bayesian network structure corresponding to:

- (a) First-order Markov process;
- (b) Second-order Markov process.



Observation Model

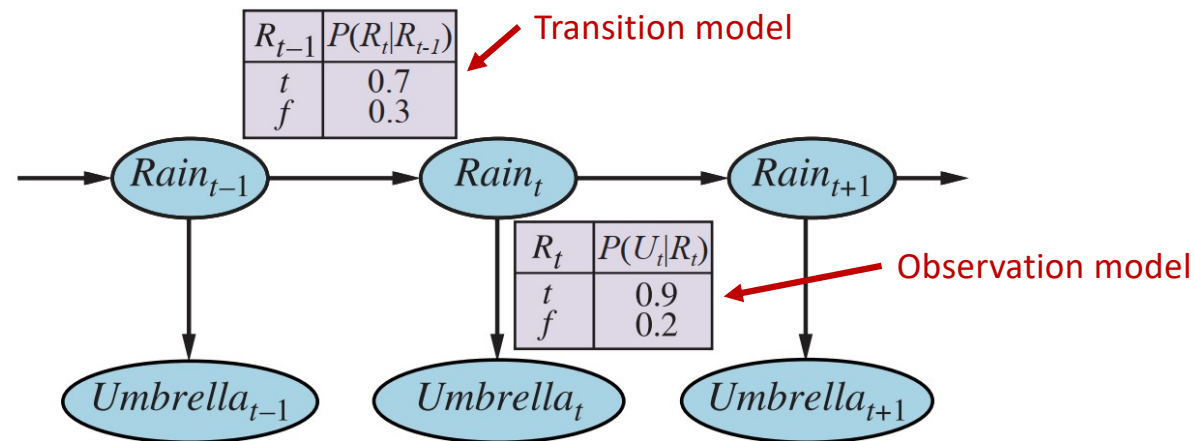
- The evidence variables E_t may, in general, depend on:
 - Previous states $X_0: X_{t-1}$
 - Previous observations $E_0: E_{t-1}$
 - As well as the current state X_t



Observation Model

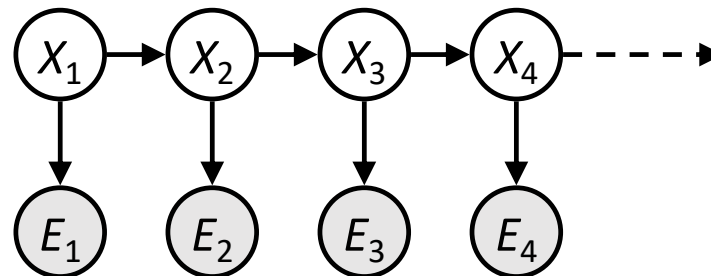
- Markov assumption for the observation model (or sensor model):
 - The current observation depends **only on the current state**, not on the full history.

$$P(E_t \mid X_0: X_{t-1}, E_1: E_{t-1}) = P(E_t \mid X_t)$$



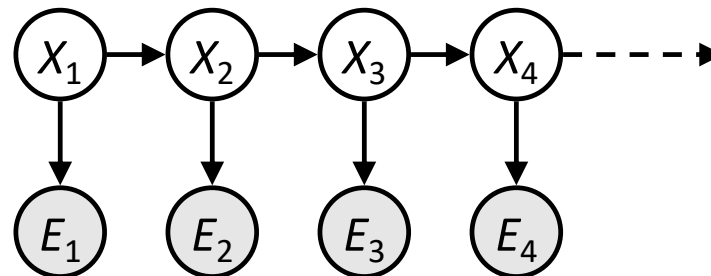
Hidden Markov Model (HMM)

- An HMM is a temporal probabilistic model where the system's **state** at each time step is represented by a **single, discrete hidden variable**.
- There is **no restriction** on the number or type of **evidence** variables.
 - They can be discrete or continuous, and multiple observations can be made at each time step.



Hidden Markov Model (HMM)

- An HMM is a temporal probabilistic model where the system's **state** at each time step is represented by a **single, discrete hidden variable**.
- What if the system has **multiple state variables**?
 - Combine them into a single “megavariable” whose values are all possible combinations (tuples) of the individual variables.



Example: Weather HMM

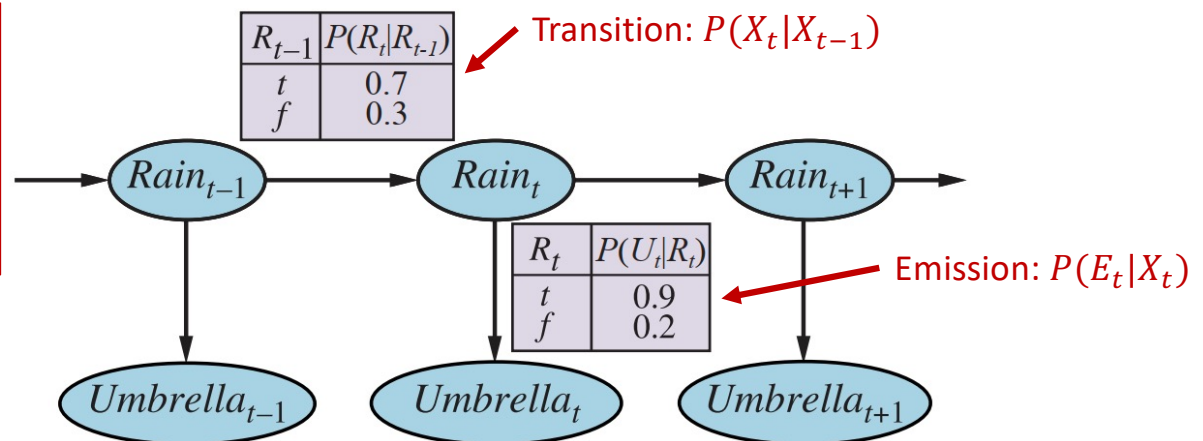
- An HMM is defined by:
 - Initial distribution: $P(X_0)$
 - Transitions: $P(X_t|X_{t-1})$
 - Emissions: $P(E_t|X_t)$

R_{t-1}	R_t	$P(R_t R_{t-1})$
+r	+r	0.7
+r	-r	0.3
-r	+r	0.3
-r	-r	0.7

R_t	U_t	$P(U_t R_t)$
+r	+u	0.9
+r	-u	0.1
-r	+u	0.2
-r	-u	0.8

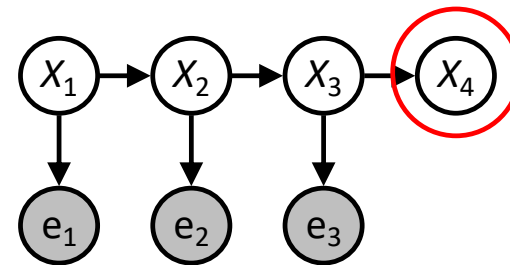
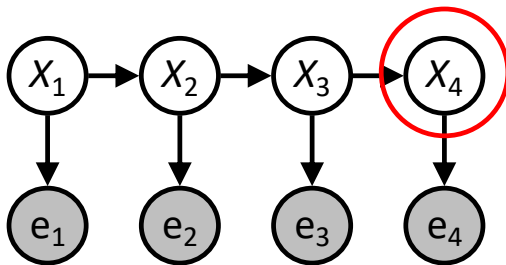
Remember two keys for HMM:

- **State:** The future state depends only on the current state.
- **Evidence:** Observations depend only on the current state.



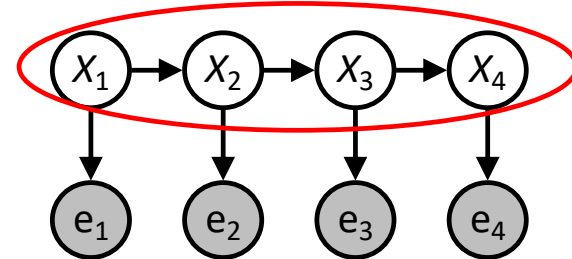
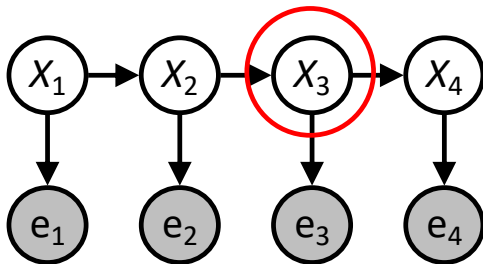
Inference Tasks

- Filtering or state estimation: $P(X_t | e_{1:t})$
- Prediction: $P(X_{t+k} | e_{1:t})$ for $k > 0$
 - Similar to filtering without evidence!



Inference Tasks

- Filtering or state estimation: $P(X_t | e_{1:t})$
- Prediction: $P(X_{t+k} | e_{1:t})$ for $k > 0$
- Smoothing: $P(X_k | e_{1:t})$ for $0 \leq k < t$
- Most likely explanation: $\arg \max_{x_{1:t}} P(x_{1:t} | e_{1:t})$



Filtering and Prediction

- $P(X_t | e_{1:t})$ - the posterior distribution over the most recent state, given all evidence up to time t .
 - Also called the belief state.
 - A rational agent maintains this belief state to make informed decisions, without needing to revisit the entire observation history.
 - Given $P(X_t | e_{1:t})$, the agent can update to the next time step using the new evidence e_{t+1} :

$$P(X_{t+1} | e_{1:t+1}) = f(e_{t+1}, P(X_t | e_{1:t}))$$

This update is performed recursively using the transition and observation models

Filtering and Prediction

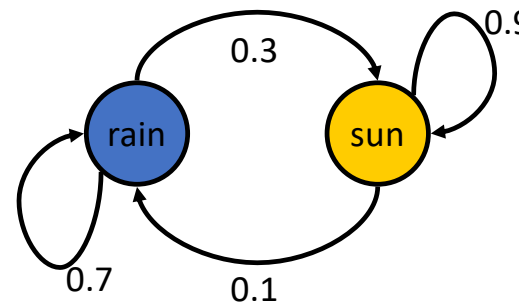
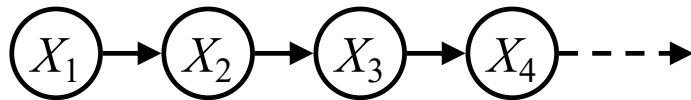
$$\begin{aligned}\mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t+1}) &= \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t}, \mathbf{e}_{t+1}) \quad (\text{dividing up the evidence}) \\ &= \alpha \mathbf{P}(\mathbf{e}_{t+1} | \mathbf{X}_{t+1}, \mathbf{e}_{1:t}) \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t}) \quad (\text{using Bayes' rule, given } \mathbf{e}_{1:t}) \\ &= \alpha \underbrace{\mathbf{P}(\mathbf{e}_{t+1} | \mathbf{X}_{t+1})}_{\text{update}} \underbrace{\mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t})}_{\text{prediction}} \quad (\text{by the sensor Markov assumption}).\end{aligned}$$



$$\begin{aligned}\mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t+1}) &= \alpha \mathbf{P}(\mathbf{e}_{t+1} | \mathbf{X}_{t+1}) \sum_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{x}_t, \mathbf{e}_{1:t}) P(\mathbf{x}_t | \mathbf{e}_{1:t}) \\ &= \alpha \underbrace{\mathbf{P}(\mathbf{e}_{t+1} | \mathbf{X}_{t+1})}_{\text{sensor model}} \sum_{\mathbf{x}_t} \underbrace{\mathbf{P}(\mathbf{X}_{t+1} | \mathbf{x}_t)}_{\text{transition model}} \underbrace{P(\mathbf{x}_t | \mathbf{e}_{1:t})}_{\text{recursion}} \quad (\text{Markov assumption}).\end{aligned}$$

Example: Forward Prediction

- What's $P(X)$ on some day t ?

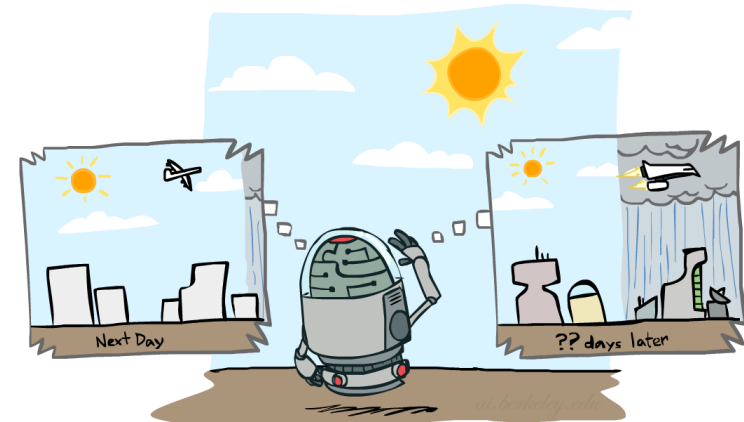


X_{t-1}	X_t	$P(X_t X_{t-1})$
sun	sun	0.9
sun	rain	0.1
rain	sun	0.3
rain	rain	0.7

$P(x_1)$ = known

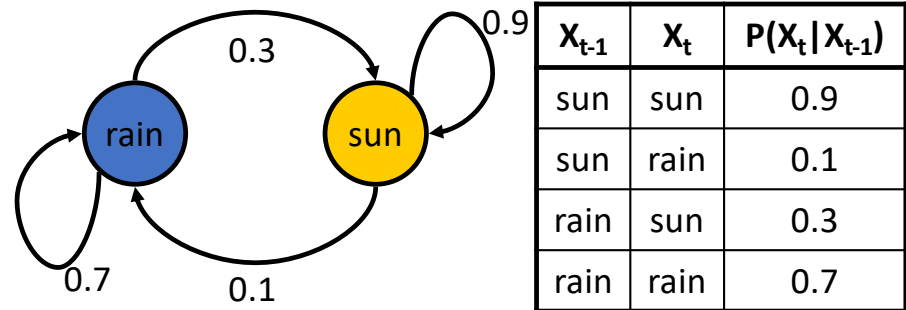
$$\begin{aligned}
 P(x_t) &= \sum_{x_{t-1}} P(x_{t-1}, x_t) \\
 &= \sum_{x_{t-1}} P(x_t \mid x_{t-1}) P(x_{t-1})
 \end{aligned}$$

Forward transition



Example: Forward Prediction

- Initial distribution: “sun”
 $P(X_1 = \text{sun}) = 1.0$

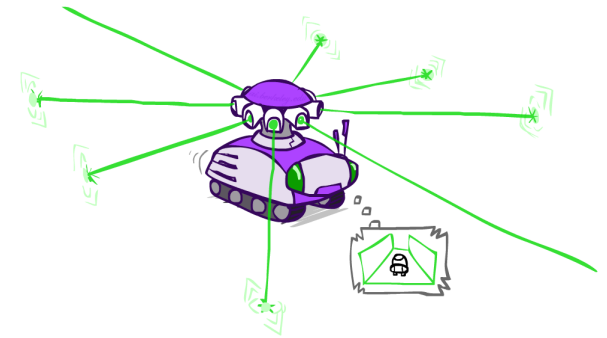
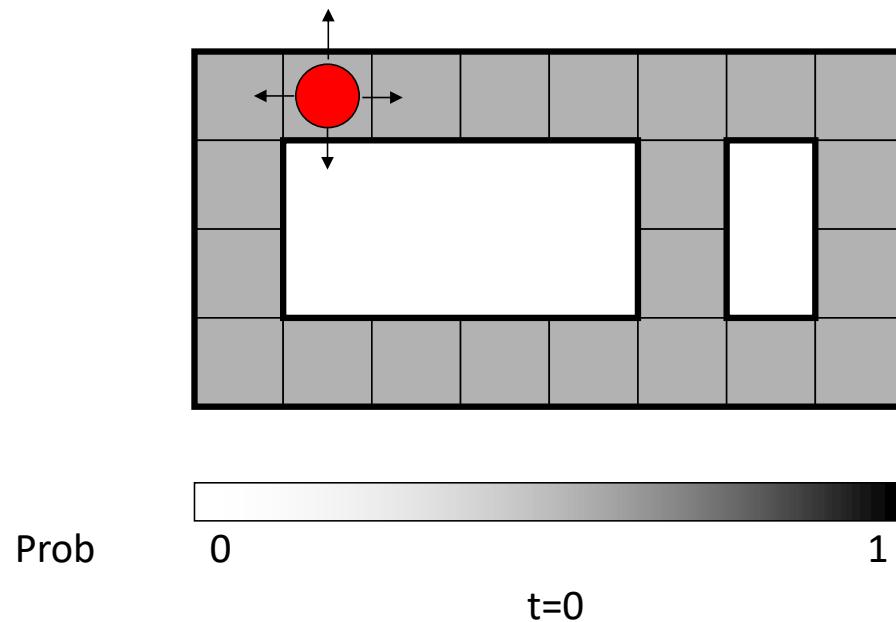


- What is the probability distribution after one step (that is $P(X_2)$)?

$$P(X_2 = \text{sun}) = \sum_{x_1} P(x_1, X_2 = \text{sun}) = \sum_{x_1} P(X_2 = \text{sun} | x_1) P(x_1)$$

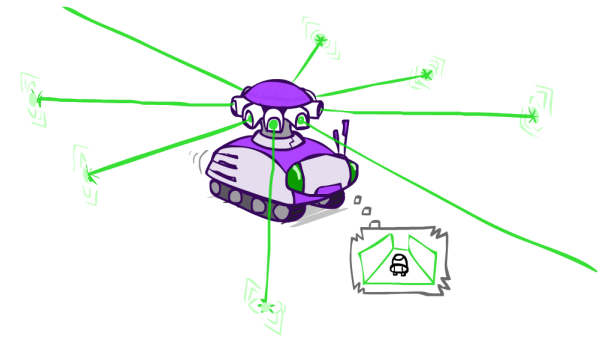
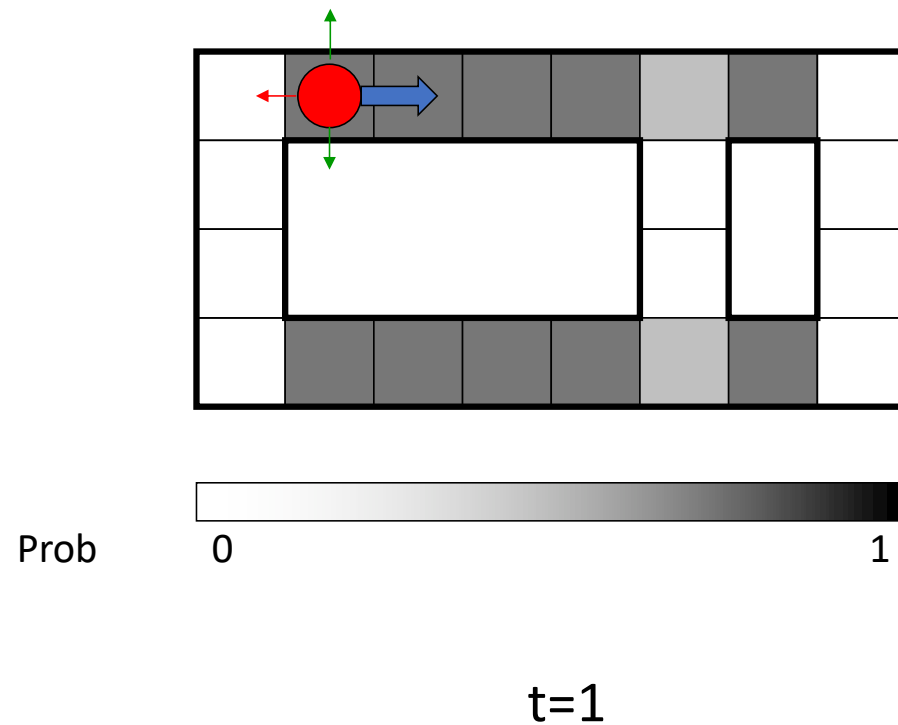
$$\begin{aligned} P(X_2 = \text{sun}) &= P(X_2 = \text{sun} | X_1 = \text{sun}) P(X_1 = \text{sun}) + \\ &\quad P(X_2 = \text{sun} | X_1 = \text{rain}) P(X_1 = \text{rain}) \\ &= 0.9 \cdot 1.0 + 0.3 \cdot 0.0 = 0.9 \end{aligned}$$

Example: Robot Localization

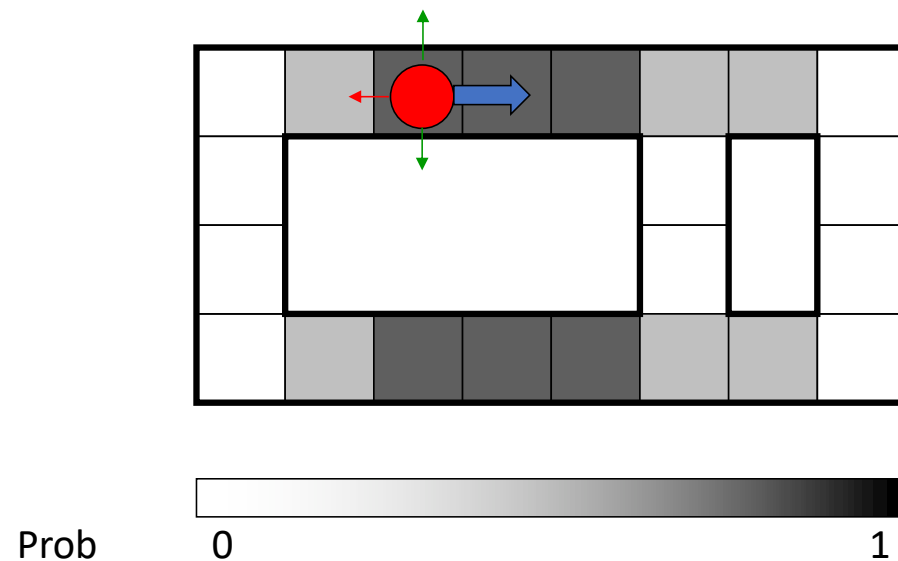


- Sensor model: can read in which directions there is a wall, never more than 1 mistake.
- Motion model: may not execute action with small probability.

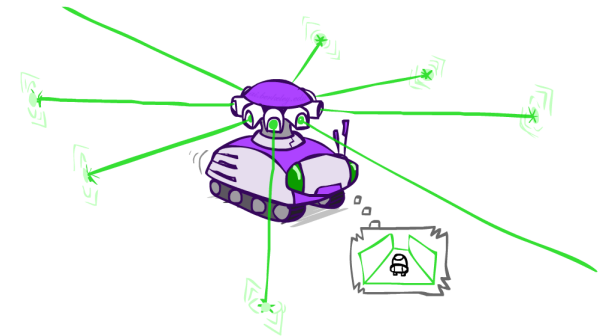
Example: Robot Localization



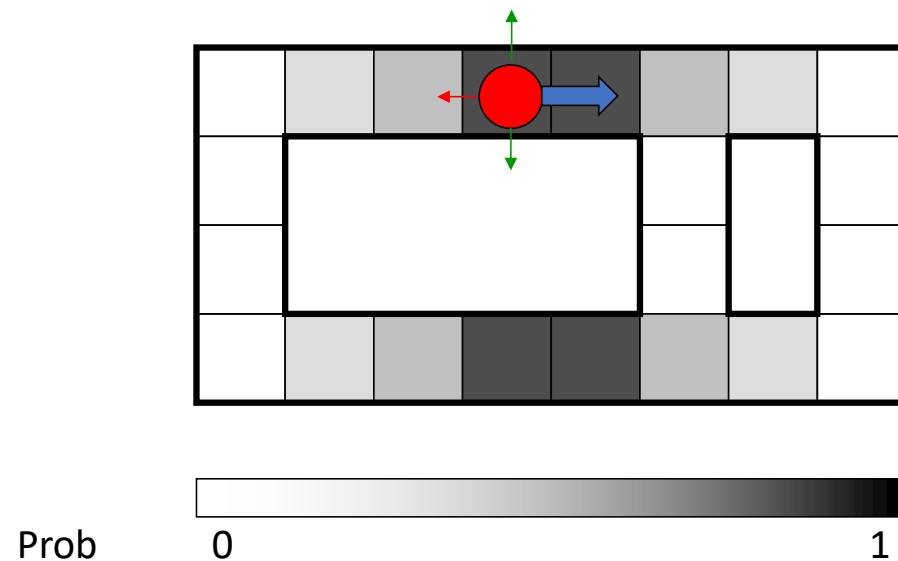
Example: Robot Localization



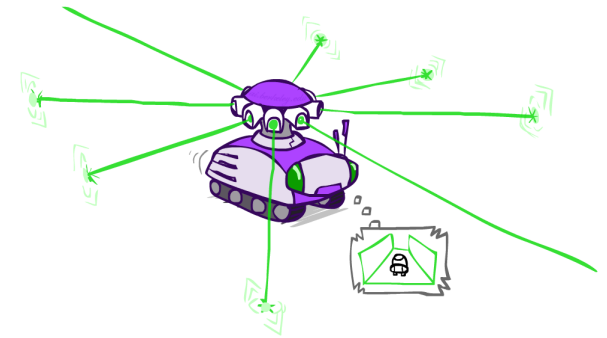
$t=2$



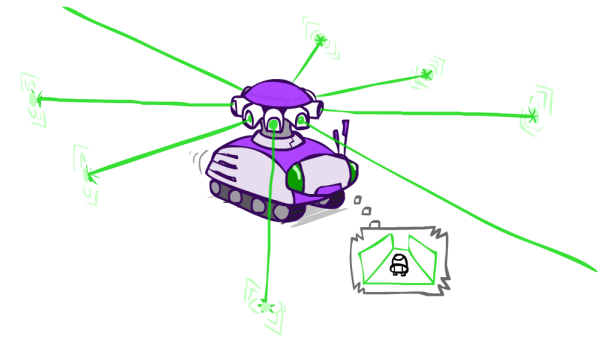
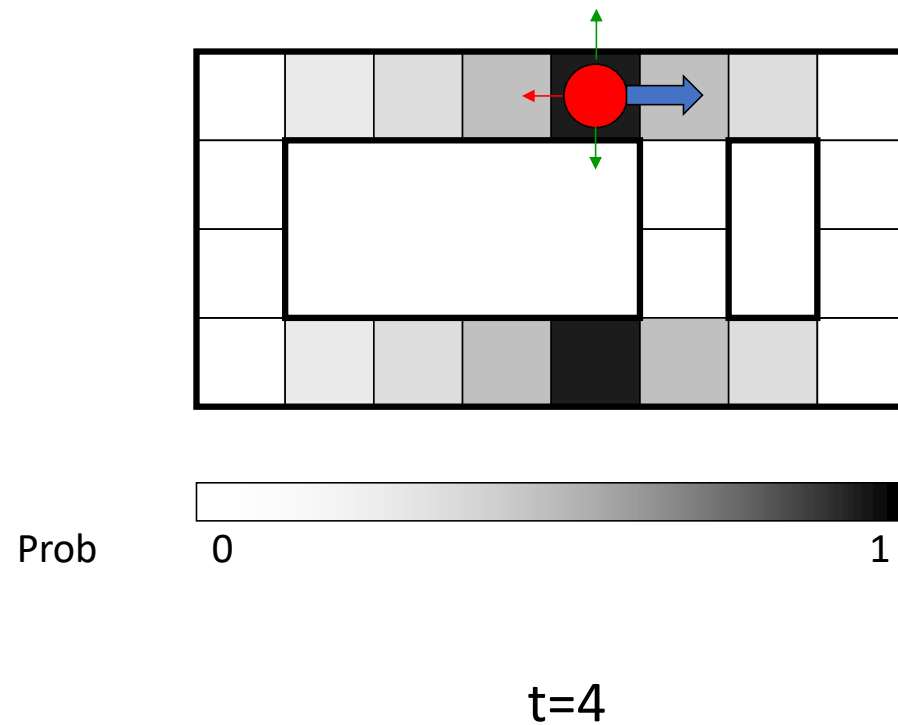
Example: Robot Localization



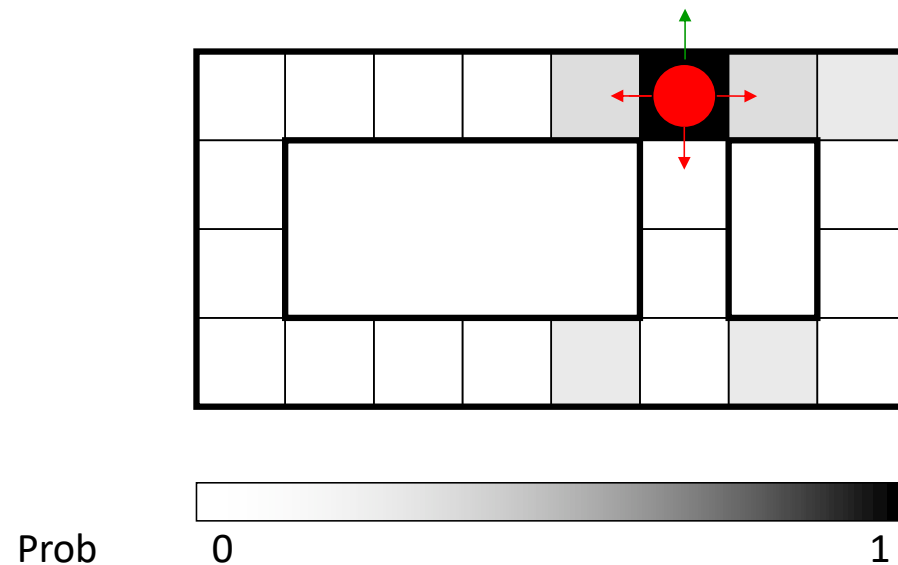
t=3



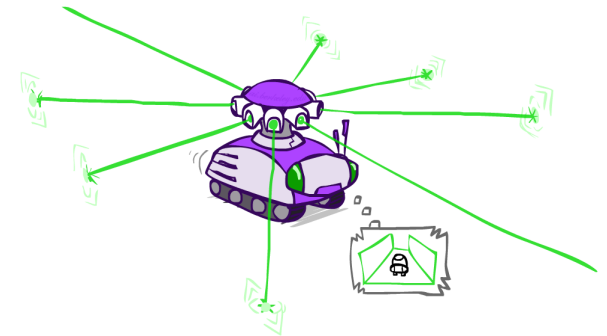
Example: Robot Localization



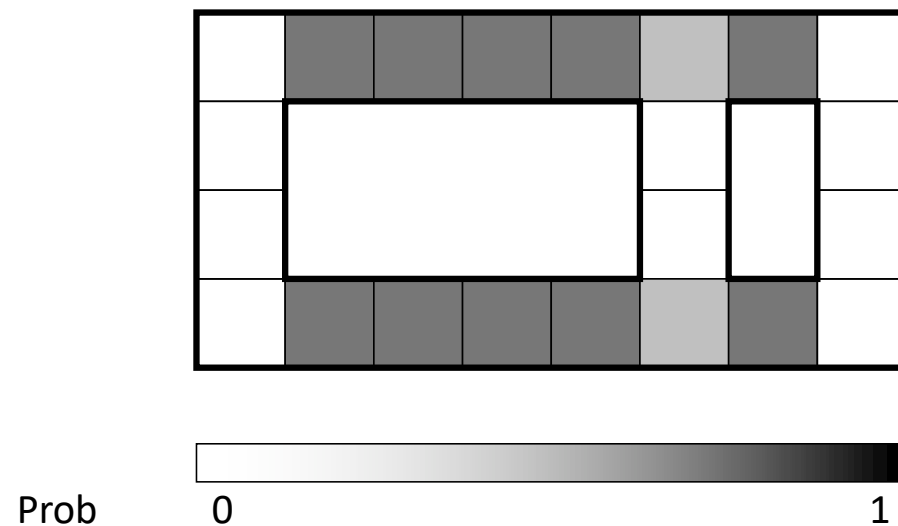
Example: Robot Localization



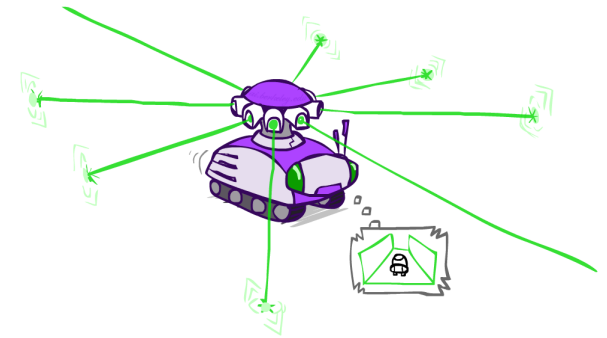
t=5



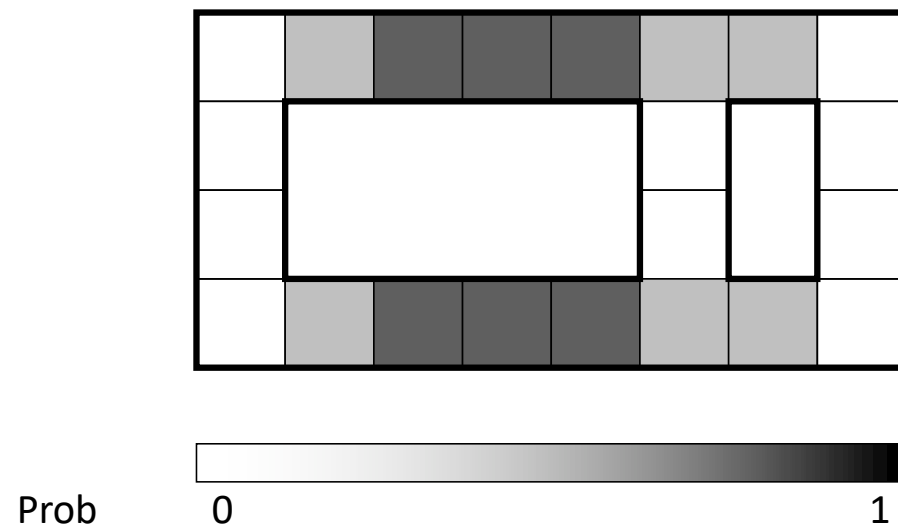
Example: Robot Localization



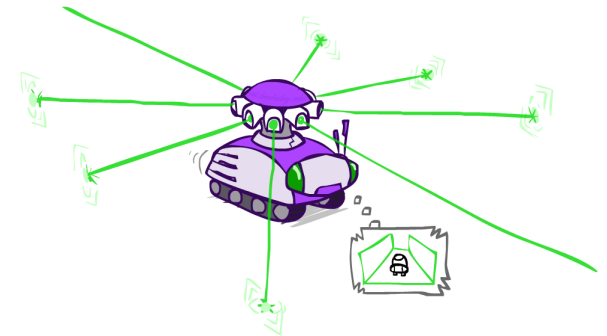
t=1



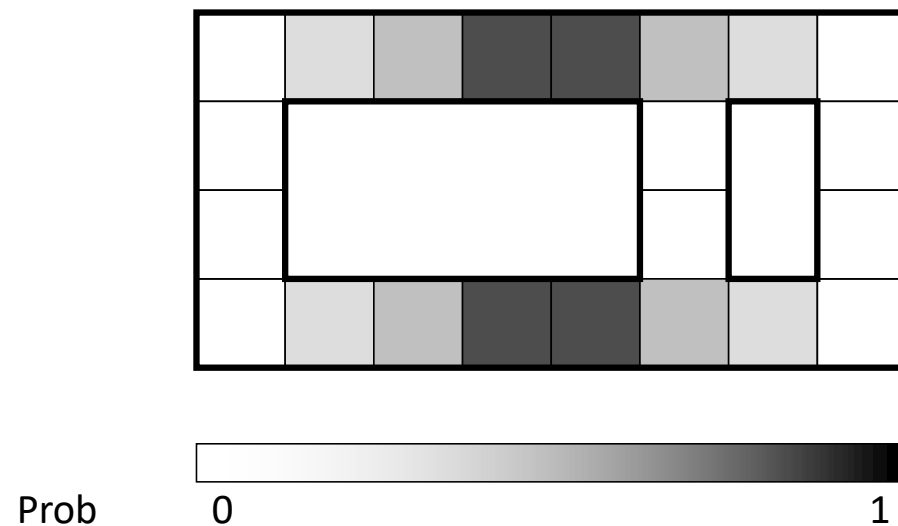
Example: Robot Localization



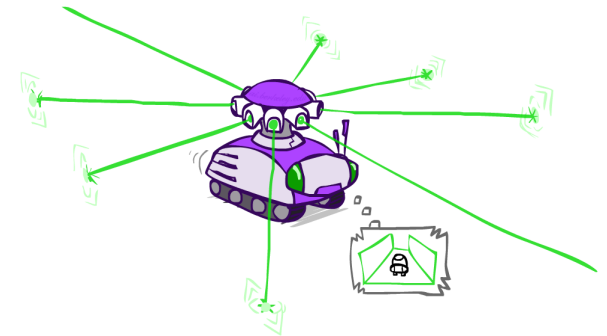
t=2



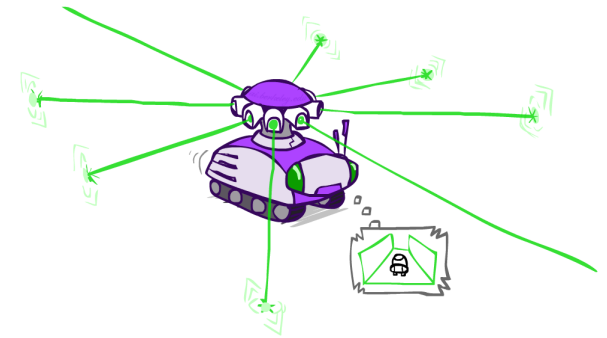
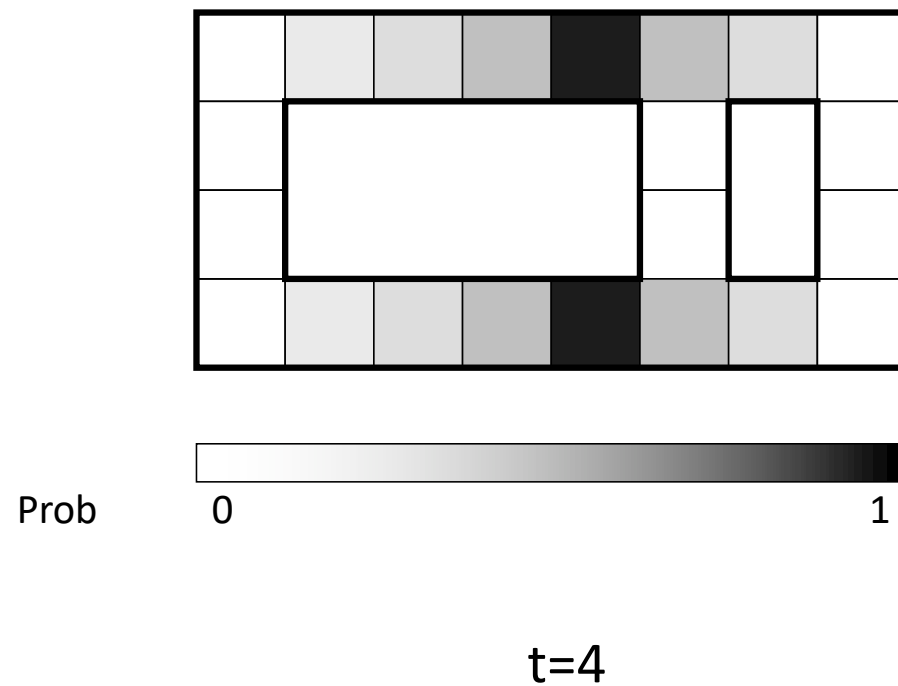
Example: Robot Localization



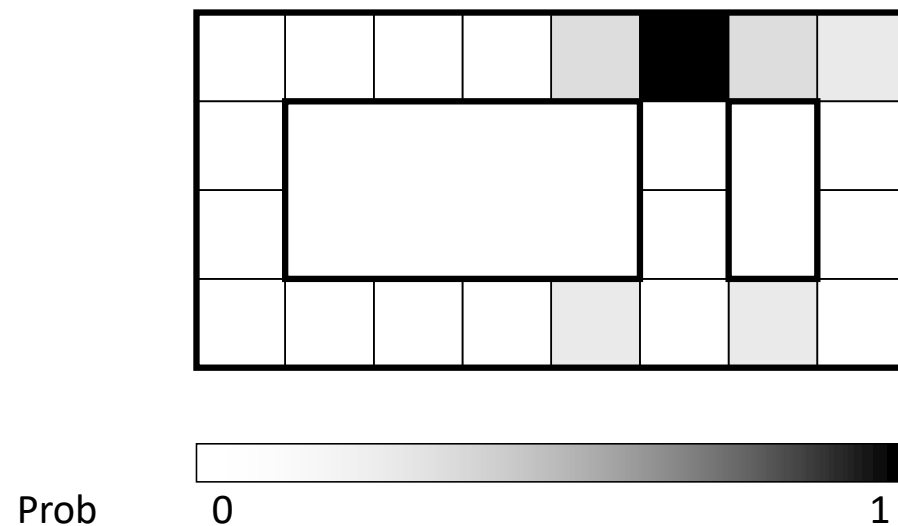
$t=3$



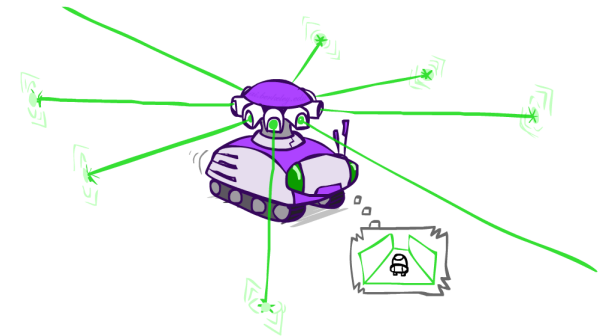
Example: Robot Localization



Example: Robot Localization



t=5



Summary: The Forward Algorithm

- We are given evidence at each time and want to know:

$$P(X_t | e_{1:t})$$

- We can derive the following updates:

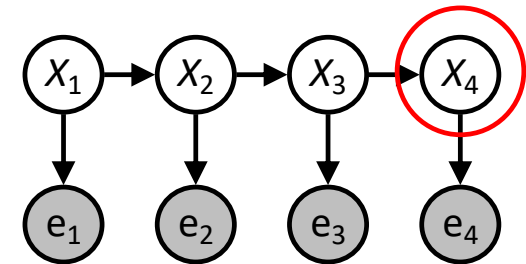
$$P(X_t | e_{1:t}) = \alpha P(e_t | X_t) \sum_{x_{t-1}} P(X_t | x_{t-1}) P(x_{t-1} | e_{1:t-1})$$

Don't forget normalization

3. Re-weight state belief at t by emissions

2. Predict state at t from $t-1$ following transitions

1. We should know state belief at $t-1$



The whole process is to propagate belief from previous state to current state, and re-weight by latest evidence observable to the agent.